CSE276 — Artificial Intelligence Foundations

Multi-Agent Systems & Human-AI Collaboration

When intelligent agents work together

Week 7 I Lecture 14 I Unit III

From One Agent to Many



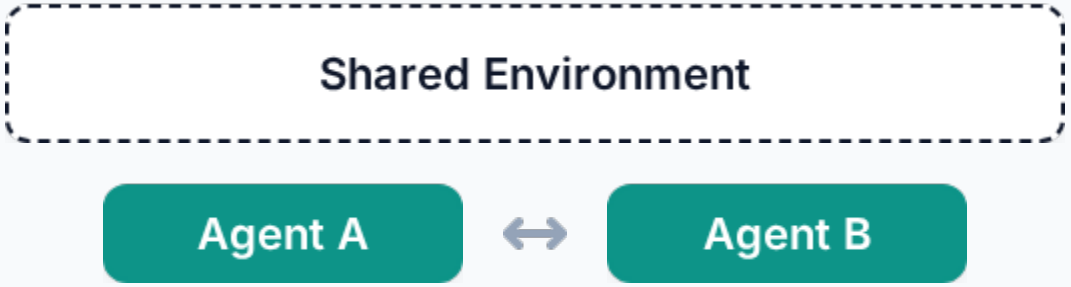
Learning Objectives

By the end of the lecture, you should be able to:

- Explain what a multi-agent system is.
- Distinguish single-agent and multi-agent environments.
- Explain cooperative and competitive agents.
- Explain why communication is important.
- Understand coordination between agents.
- Understand basic negotiation & conflict concepts.
- Explain human-AI collaboration.
- Identify real-world applications of MAS.
- Connect these concepts to **Practical 6**.

What Is a Multi-Agent System?

A **multi-agent system** is a system in which multiple agents interact within a shared environment.





- Each agent perceives, decides, and acts.
- Agents may cooperate, compete, or simply interact.

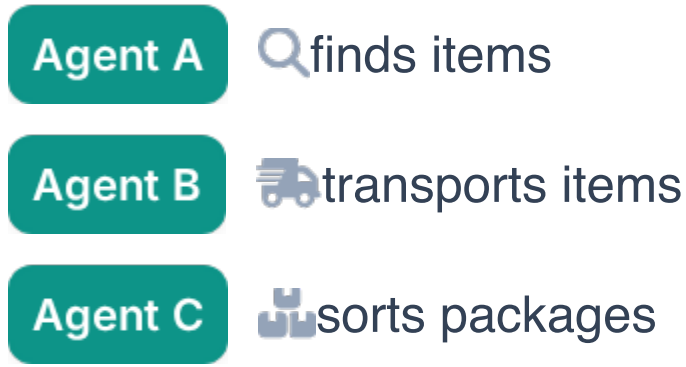
Why Use Multiple Agents?

Question: "Why not just build one powerful agent?"

Advantages:

- **Divide** a large problem.
- Work in **parallel**.
- Handle different types of tasks.
- Improve system **robustness** (no single point of failure).
- Operate in different physical locations.
- Allow **specialization**.

Example: Warehouse



Single-Agent vs Multi-Agent

Single-Agent

One primary decision-maker.

Example:
Crossword-solving AI

Multi-Agent

Multiple interacting decision-makers.

Examples
:

- Chess
- Warehouse robots
- Autonomous vehicles
- Drone teams

Cooperative vs Competitive



Cooperative

Agents work toward compatible or shared goals.

Examples :

- Warehouse robots
- Search-and-rescue drones
- Robot teams (e.g., RoboCup)

Competitive

Agents have conflicting goals.

Examples :

- Competitive video games
- Chess / Poker AI
- Auction/bidding scenarios

Note: Some complex environments contain both cooperation and competition simultaneously.

Why Do Agents Need Communication?

Warehouse scenario: **Agent A finds an obstacle.**

Question: Should Agent A tell Agent B?

✕ **Without communication:**

Agent B may unknowingly choose the same blocked route and waste time.

✓ **With communication:**

Agent B receives the update and instantly calculates an alternate route to avoid the obstacle.

> **Communication** allows agents to share information that can improve overall decision-making.



What Can Agents Communicate?



Location

"I am at grid X:4, Y:12"



Status

"My battery is at 15%"



Intentions

"I am heading to Aisle 4"



Warnings

"Aisle 3 is blocked!"



"Aisle 3 is blocked."

Robot A





"Understood. I'll use Aisle 4."

Robot B

Communication Does Not Mean Cooperation

Important Distinction

Scenario: Two competing chess agents.

They communicate: *"I am moving my queen to D4."*

Would that necessarily help both agents?

No.

- Agents can communicate even when their goals conflict.

- **Communication** is simply the *exchange of information*.
- **Cooperation** is *working toward compatible/shared goals*.



What Is Coordination?

Coordination means organizing the actions of multiple agents so that their actions work well together.

Example: 3 Warehouse Robots

Without coordination:
Robot A → aisle 1
Robot B → aisle 1
Robot C → aisle 1
Result: Traffic jam.

With coordination:
Robot A → aisle 1
Robot B → aisle 2
Robot C → aisle 3
Result: Smooth performance.



Communication vs Coordination



Communication

Sharing information

> “Aisle 1 is blocked.”



Coordination

Organizing actions

> “You take Aisle 2; I'll take Aisle 3.”

Communication can **support** coordination, but communication and coordination are **not** the same thing.

Task Allocation

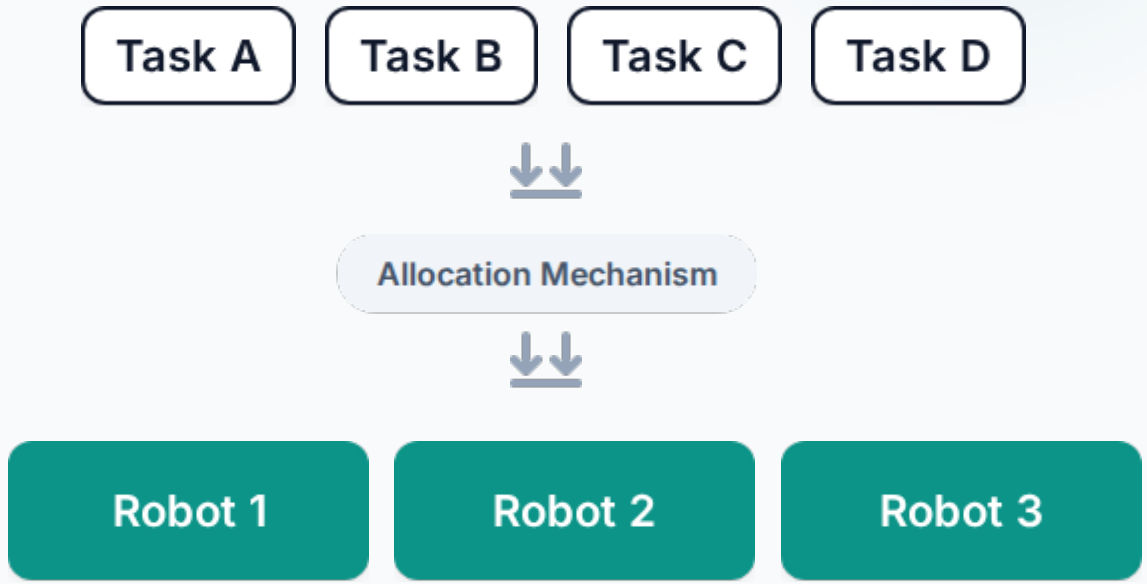
A common coordination problem.

Scenario: 4 delivery tasks. 3 robots.

Question: Who should do which task?

Factors to consider:

- Distance to target
- Battery level
- Availability
- Skill/capability
- Priority of the task



What Happens Without Coordination?



Without Coordination

Two agents.

Both choose the same task.

↓

Duplicate work.
Another task remains unfinished.



With Coordination

Agent A → Task 1

Agent B → Task 2

↓

Better coverage.
Optimal use of resources.

What If Agents Want Different Things?

Scenario: Two delivery robots want the same charging station.

Agent A

Battery = 10%

Agent B

Battery = 40%

Question: Who should use the charger first?

Introducing: **Conflict Resolution / Negotiation**

When agents have competing needs, they need rules or mechanisms for deciding who gets what.



Simple Negotiation

Two agents want the same resource. How do they decide?

Priority Rules

Pre-defined logic (e.g., lower battery goes first).

First-Come

Whoever arrives at the resource first claims it.

Auction / Bidding

Agents 'bid' for the resource based on urgency.

Shared Scheduling

Dividing time (A takes 10 mins, then B takes 10 mins).

AI Alone vs Human + AI

Question: Does AI always need to replace the human?



AI Alone

AI makes the decision entirely autonomously.



Human Alone

Human makes the decision without AI assistance.



Human + AI

Human and AI contribute different, complementary strengths.

Collaboration can combine human judgment with machine capabilities.

What Are Humans Good At?

Humans can provide...

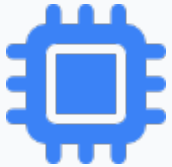
- Contextual judgment
- Common sense reasoning
- Ethical and moral reasoning
- Understanding unusual/novel situations
- Empathy and emotional intelligence
- Navigation of highly ambiguous decisions



The Human Edge

While AI excels in structured environments, human beings possess a broad understanding of the world that allows them to make sensible decisions when rules break down or data is missing.

What Can AI Contribute?



AI can provide...

- Extremely fast data processing
- Complex pattern detection
- Large-scale statistical analysis
- Consistent, repetitive processing (no fatigue)
- Data-driven predictions & recommendations

Human Strengths

+

AI Strengths

=

Collaborative Decision-Making

Human-in-the-Loop

Human-in-the-loop (HITL) means a human remains involved in the decision, supervision, or control process.





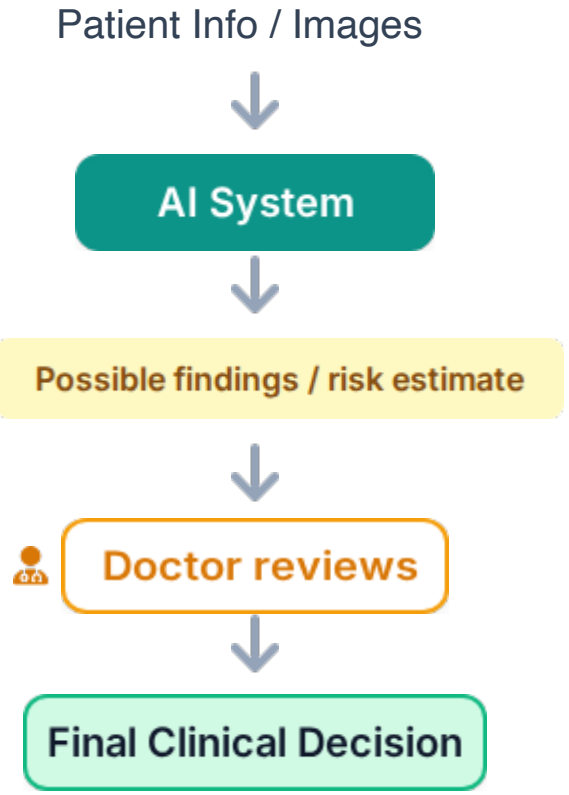
Examples:

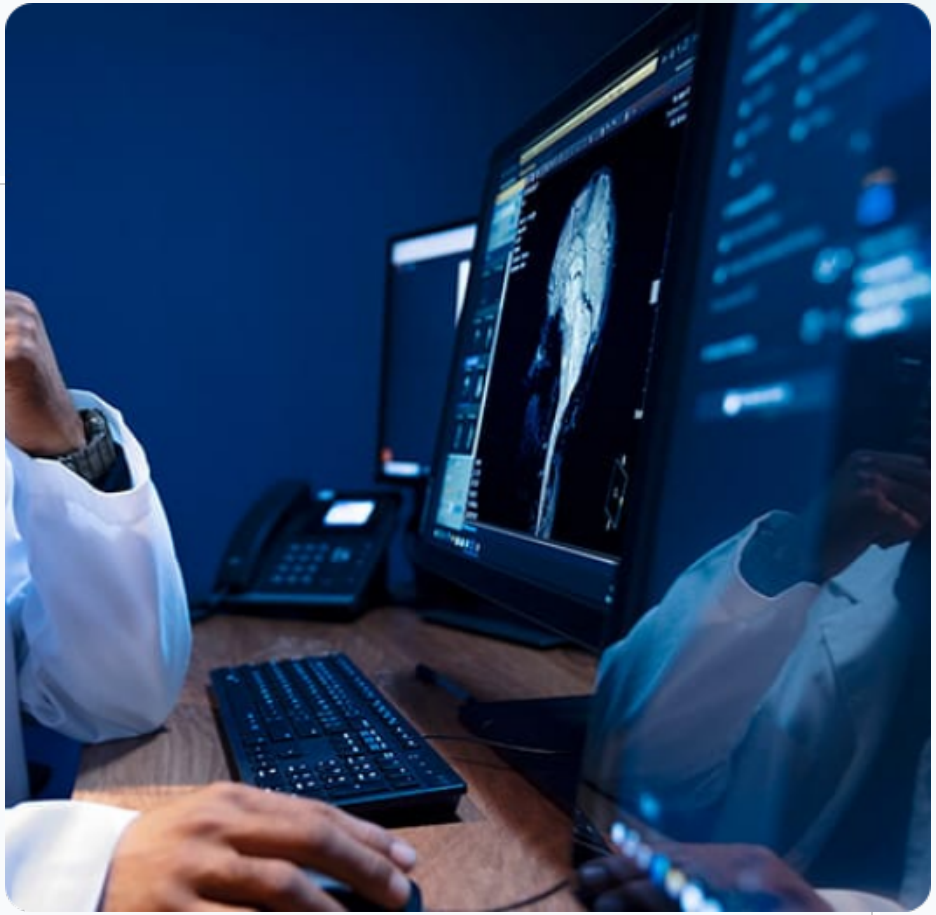
- Medical decision support
- Fraud detection approval
- Content moderation

The exact role of the human depends on the application and risk level.

Human-AI Collaboration Example

Medical Diagnosis





AI can **support** decision-making without being the final decision-maker.

Multi-Agent Systems in the Real World



Warehouse Robotics

Multiple robots coordinate movement and tasks.



Autonomous Vehicles

Vehicles interact within traffic safely.



Drone Teams

Agents divide search and mapping areas.



Smart Grids

Distributed systems coordinate energy resources.



Multiplayer Games

Agents cooperate or compete in complex spaces.



Search & Rescue

Robots coordinate exploration in disaster zones.

Practical 6 — Multi-Agent Collaboration

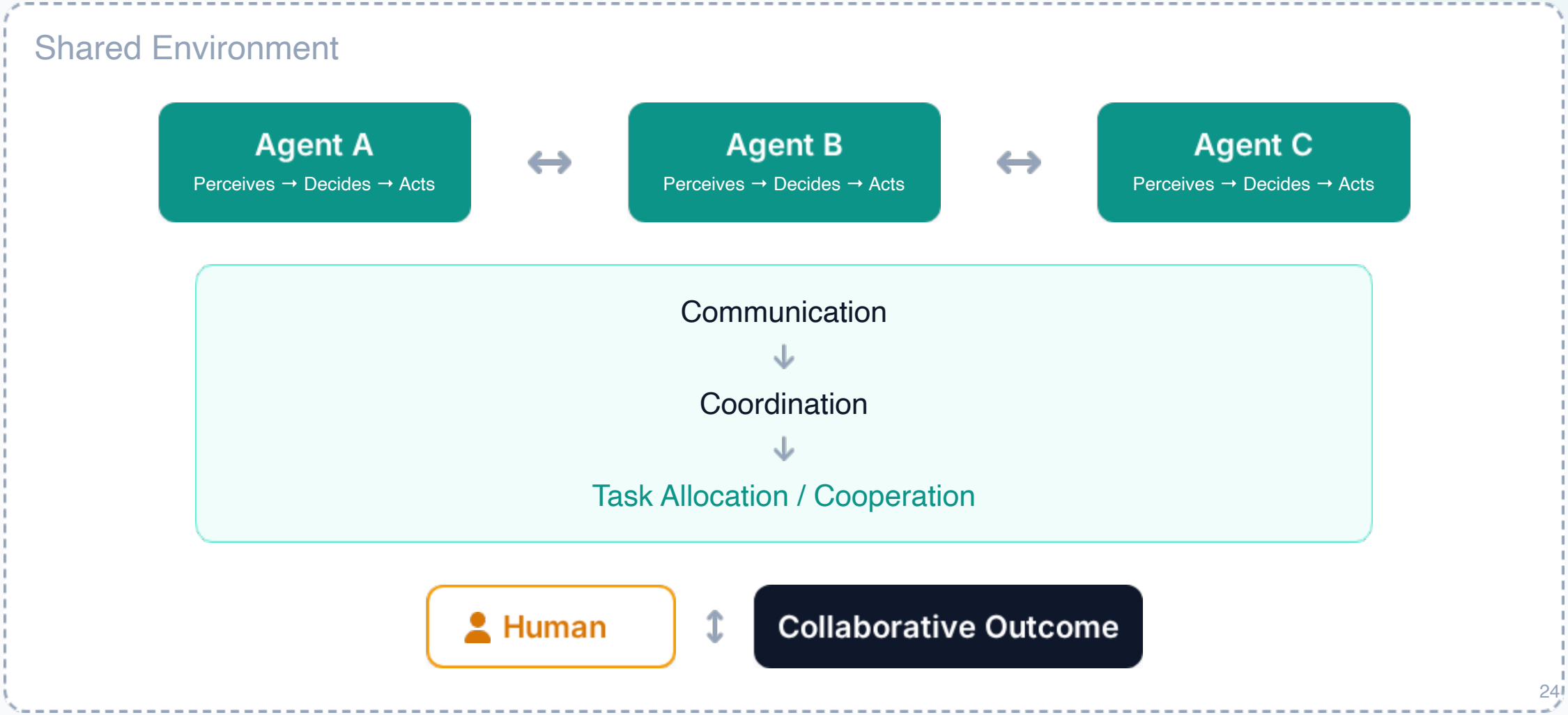
Conceptual Workflow

1. Define environment
2. Define agents
3. Define each agent's goal
4. Define communication
5. Define coordination
6. Run interaction
7. Evaluate system performance

What you should identify:

- Who are the **agents**?
- What is the shared **environment**?
- What are the **goals**? (Coop vs Comp)
- How does **communication** happen?
- How is **coordination** achieved?
- Are there **conflicts**?
- Is there **human involvement**?

Putting Everything Together



Final Recap

Multi-Agent System: Multiple interacting agents.

Communication: Share information.

Coordination: Organize actions.

Cooperation: Work toward compatible goals.

Competition: Goals may conflict.

Human-AI Collab: Combine human & AI strengths.

Many agents + communication + coordination
=
Collective problem solving

Classroom Question

Imagine three delivery robots working in the same warehouse. What could go wrong if they cannot communicate or coordinate? How would you design a simple solution?

Next Unit:

Machine Learning & Data-Driven AI

Image Sources



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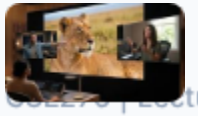
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